Growth, Satisfaction, and Risk: An Interactive Business Intelligence Analysis of E-Commerce Performance

Project Report

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1. Introduction

1.1 Narrative

Growing an e-commerce business is an exercise in managing friction. The drive for higher revenue often cracks the underlying infrastructure, leading to a collapse in service quality that management ignores as long as targets are met. This report investigates exactly where that breakdown is happening. The objective is to determine how a business can expand its financial footprint without hitting a satisfaction ceiling that eventually leads to brand erosion.

A heavy reliance on high average star ratings is a significant pitfall in modern e-commerce management. As Kim et al. (2025) suggest, aggregate scores are often misleading because they mask the specific service attributes that realistically influence customer sentiment. To address operational failures before they result in lost revenue, it is necessary to move beyond surface level metrics and analyze the specific details within those reviews. Marketing strategies require a similar level of analytical depth. Generic, broad tactics are increasingly ineffective in a saturated e-commerce market. Ghosh and Chatterjee (2025) argue that the analysis of point-of-sale (POS) big data is the most reliable method for moving past assumptions to understand actual shopper behavior. By evaluating real transactional data, a business can develop customer segments that reflect the true complexity of the market.

These review patterns essentially function as the bridge between a browser and a final transaction. Zhao et al. (2021) demonstrated that online reviews act as a critical trust signal with a direct impact on revenue growth and consumer intention. If a high-earning product begins to receive negative feedback, it creates an immediate and measurable risk for the entire company. By using Tableau here, it serves as the visualization tool for mapping these intersection points. The report uses this capability to replace assumption with precision in identifying where business intervention becomes necessary. Strategy formation then proceeds from actual data patterns rather than from projections based on optimism.

1.2 Research Question

How do product mix, customer demographics, and purchasing behaviours influence revenue performance and customer satisfaction, and how can the retailer optimize growth while mitigating the risk of low review scores?

Sub Questions (This is to address the broader research question**)**

1. Which product categories and cities generate the most revenue, and are there specific payment methods or seasonal trends that drive these totals? (Revenue Performance)
2. How do different age groups and genders vary in their average order values and how often do they return to the store? (Demographic)
3. Which products or categories show a pattern of high revenue combined with low ratings, and what can be done to fix these risks before they affect growth? (Customer Satisfaction & Risk)

1.3 Hypothesis

H1: Annual sales peak during major seasonal shopping windows and holiday breaks rather than staying consistent all year.

H2: Older age groups and specific genders spend more on average per order, identifying the primary revenue drivers for the business.

H3: High sales volumes affect review scores because service quality falls behind as a product category scales up.

1.4 Data Origin and Collection

The analysis utilizes a synthetic retail dataset from Kaggle, which was programmatically created to simulate the transactional logs of a real-world e-commerce point-of-sale system. This data was selected because it preserves the essential mathematical logic of e-commerce operations, specifically the relationship between quantity and unit price, while avoiding the privacy constraints associated with actual customer files. Utilizing this structured environment allows for the development of high-precision diagnostic dashboards that identify exactly how scaling specific product categories might impact customer satisfaction and long-term brand equity.

Dataset Link: https://www.kaggle.com/datasets/ertugrulesol/online-retail-data/data

1.5 The Data Breakdown

This dataset has 10,000 rows and 13 columns. It covers different aspects from the order date to the final rating.

1. Total Rows: 10,000
2. Total Variables: 13

Dimension: 10,000 observations x 13 variables.

1.6 Data Dictionary:

Variable Name	Variable Description
customer_id	A unique numerical identifier assigned to each registered customer.
order_date	The specific calendar date when the purchase was recorded in the system.
product_id	A unique numerical identifier that is used to track each specific product item.
category_id	A numerical code representing the broader classification or group of the product.
category_name	The descriptive name of the product category, such as Electronics or Fashion.
product_name	The specific commercial or brand name of the individual item sold.
quantity	The total number of units of a product purchased within a single transaction.
price	The unit price of the product at the specific time of the transaction.
payment_method	The financial channel used to complete the purchase, such as a Credit Card or PayPal.
city	The geographic city where the customer is located.
review_score	A customer-provided satisfaction rating measured on a scale of 1.0 to 5.0.
gender	The identified gender of the customer making the purchase.
age	The numerical age of the customer at the time of the transaction.

2. Analysis, Storytelling and Visualization

2.1 Analytical Goal and Storyline

The purpose of this analysis was to provide answers to a practical business question: how does a company increase revenues and at the same time maintain customer satisfaction? By taking both customer performance and customer satisfaction into consideration, a series of dashboards have been developed that illustrate the source of revenues, which client segments drive revenue, and where low-rated customers may endanger future revenue growth. As such, the story is organized around a specific decision-making process:

1. What is the overall health of the business?
2. Where exactly is revenue being created?
3. What experience risks are hidden behind revenue?

This is why the output is structured into three connected dashboards:

Overview → Revenue → Customer Satisfaction.

2.2 Dataset and Core Measures

The dataset contains transactional records with customer and product context:

- **Customer**: customer_id, age, gender, city
- **Transaction**: order_date, quantity, price, payment_method
- **Product**: product_name, category_name
- **Experience signal**: review_score (star rating)

To make the dataset "analysis-ready" in Tableau, the key measures were engineered to translate raw fields into decision metrics:

Revenue (sales value)

- Built from the transaction level to represent value created:
- Revenue = quantity × price

Customer count

- Used as a business-facing metric (unique customers in scope):

- Customers = COUNTD(customer_id)

Average Order Value (AOV)

- Used to represent basket value.
- In practice this is computed as total revenue divided by the order key used in the workbook.

Average Rating

- Summary customer experience indicator:
- Avg Rating = AVG(review_score)

Alongside these KPIs, the analysis repeatedly uses ranking (Top N) and segmentation (age bands, gender, city, category, payment method) to surface patterns that can be acted on.

2.3 Why This Approach Was Chosen

A single model or a single chart cannot answer a multi-part business question. The approach was therefore designed as a layered diagnostic, combining:

- **Trend analysis** to reveal seasonality and momentum.
- **Contribution analysis** to show which factors affect the business.
- **Segmentation analysis** to show heterogeneity across customer groups like age, gender and city.
- **Risk framing** to connect experience to business impact. (risk–revenue trade-offs)

This analysis uses a three-layered approach to test the hypotheses. Univariate analysis establishes the baseline for revenue and ratings. Bivariate charts then isolate specific growth drivers like city and category performance. Finally, multivariate heatmaps reveal the hidden risks where age and category interact to hurt satisfaction. This structure ensures the findings move past surface-level averages to find real operational failures

2.4 Visualization Design and Storytelling Output

2.4.1 Dashboard 1: Overview

Purpose: To give an initial quantitative means to assess business performance, and identify main value pools before performing a more detailed drill down analysis.

Figure 1: Dashboard 1

- **Headline KPIs (Revenue, Customers, Average Order Value, Average Rating):** Create a baseline for future comparison and place all other data in the correct context. Total revenue (~737K $), customer count (1,000), AOV (~737 $), and average rating (~4.0 stars) inferring a financially successful company that has comfortable customer satisfaction.

Figure 2: Headline KPIs

- **Growth chart (monthly bars):** communicates seasonality and identifies peak vs weak months. This isn't just "what happened"; it sets up the business question: are peaks driven by demand, product mix, or promotions, and can they be repeated?

Figure 3: *Growth Overview*

- **Product Sales tree map:** shows the revenue distribution across categories so management sees concentration risk and opportunity. The view indicates that revenue is relatively distributed across major categories like Electronics, Sports & Outdoors, Books & Stationery, Home & Living, Fashion. It implies growth can come from multiple levers rather than one dependency.

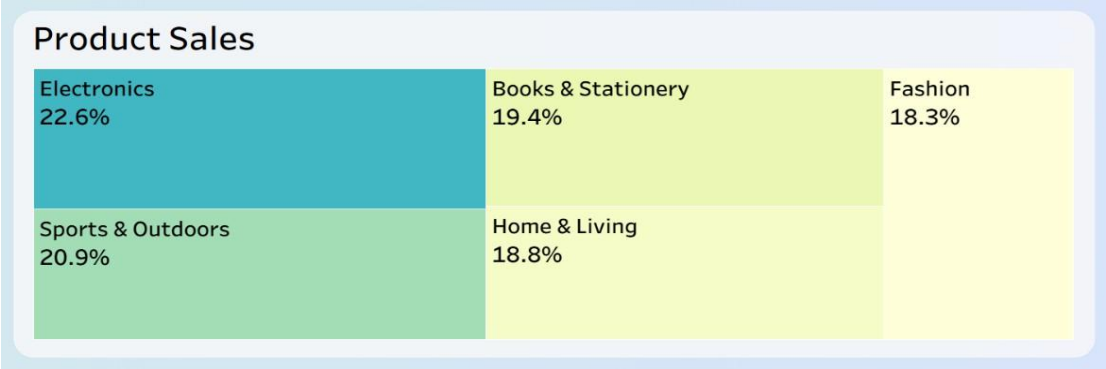


Figure 4: *Product Category Contribution in Revenue*

- **Top N revenue cities:** identifies geographic concentration. This is useful operationally to understand regional campaigns, stocking strategy, delivery partnerships and commercially to know where demand already exists.

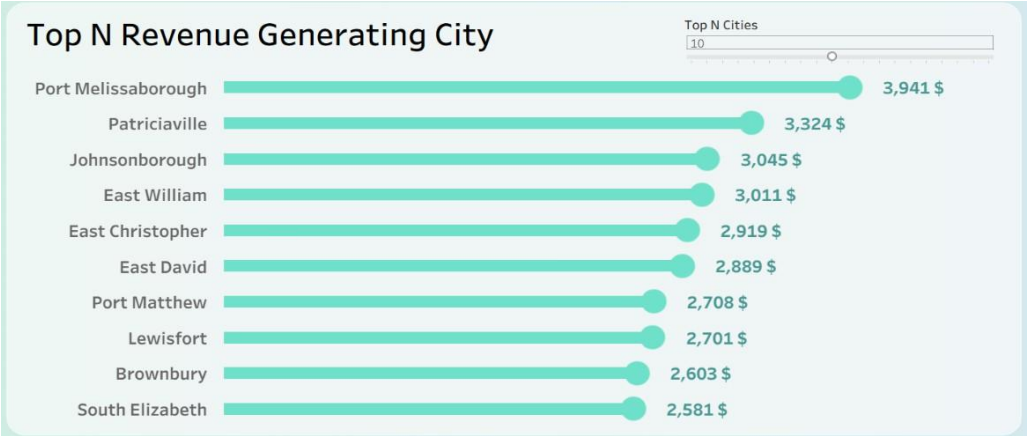


Figure 5: *Top Revenue Generating Cities*

- **Customer segmentation rings (Gender + age groups):** links who the customers are to where revenue is being generated. The intention is not demographic reporting; it is to identify which segments are worth tailoring strategy for.

Figure 6: *Customer Segmentation*

Outcome from Overview dashboard:

The business has stable overall performance, multiple category value pools, and clear geographic and demographic structure. Those were enough evidence to justify drilling into drivers (Revenue dashboard) and risks (Customer Satisfaction dashboard).

2.4.2 Dashboard 2: Revenue

Purpose: To provide a revenue timeline and insight into controllable levers to help with future growth of revenue.

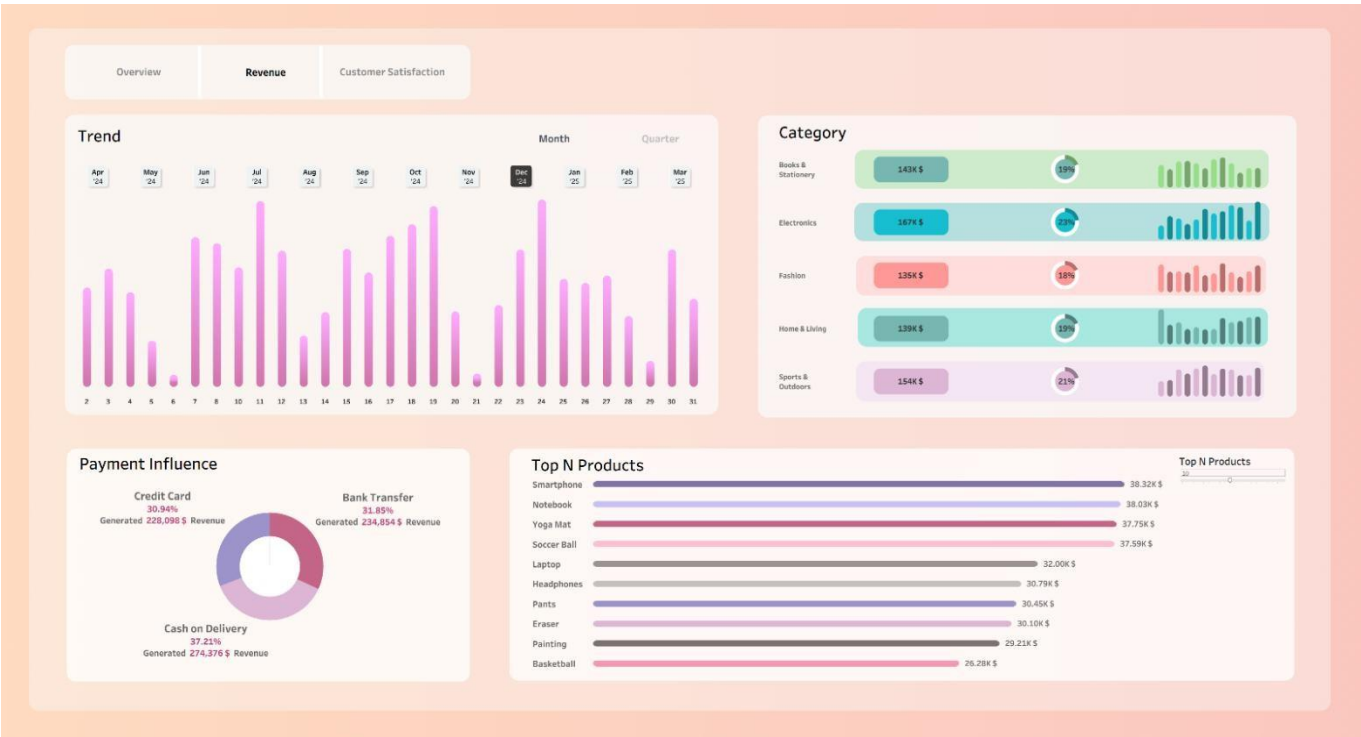


Figure 7: *Dashboard 2*

- **Trend view with time controls (Month/Quarter):** allows switching the granularity to check whether performance is driven by short promotions or consistent demand. This supports decision making on campaign timing and inventory planning.

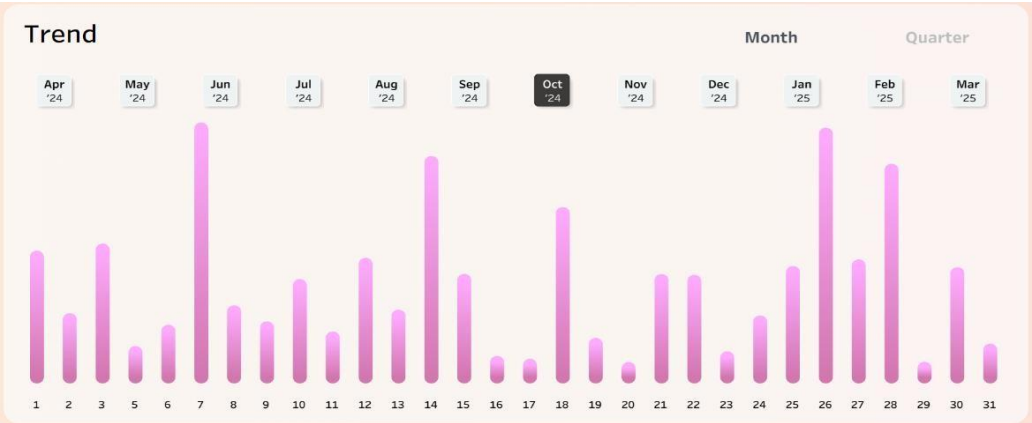


Figure 8: *Monthly and Quarterly Revenue Trend*

- **Category performance panel:** combines revenue amount, contribution share, and micro-trends. This turns 'category ranking' into a management tool:
    - High revenue + rising trend → Protect and scale
    - Mid revenue + rising trend → Invest
    - High revenue + declining trend → Diagnose pricing, competition and fatigue.

Figure 9: *Category Performance Panel*

- **Payment Influence donut:** treats payment method as a revenue lever and a friction indicator. If one method generates high revenue share, it becomes a priority for user

experience reliability, checkout optimization, and promotions. It also sets up the satisfaction story.

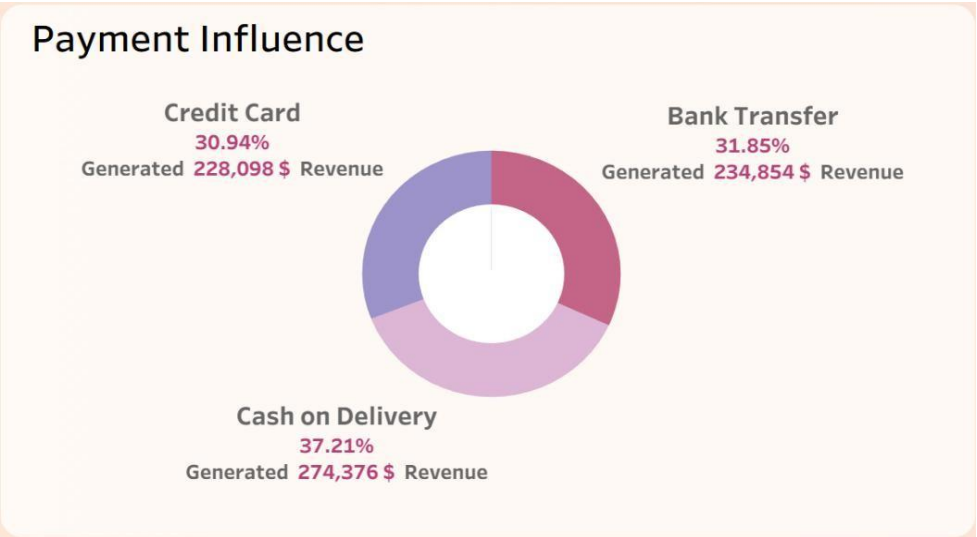


Figure 10: *Influence of Different Payment Methods on Revenue*

- **Top N Products bar chart (with parameter):** isolates the product-level revenue engine. The parameter-based Top N keeps the view decision-ready: management can shift from 'Top 10' to 'Top 20' to understand whether revenue is concentrated in a few hero products or distributed across a broader mix.

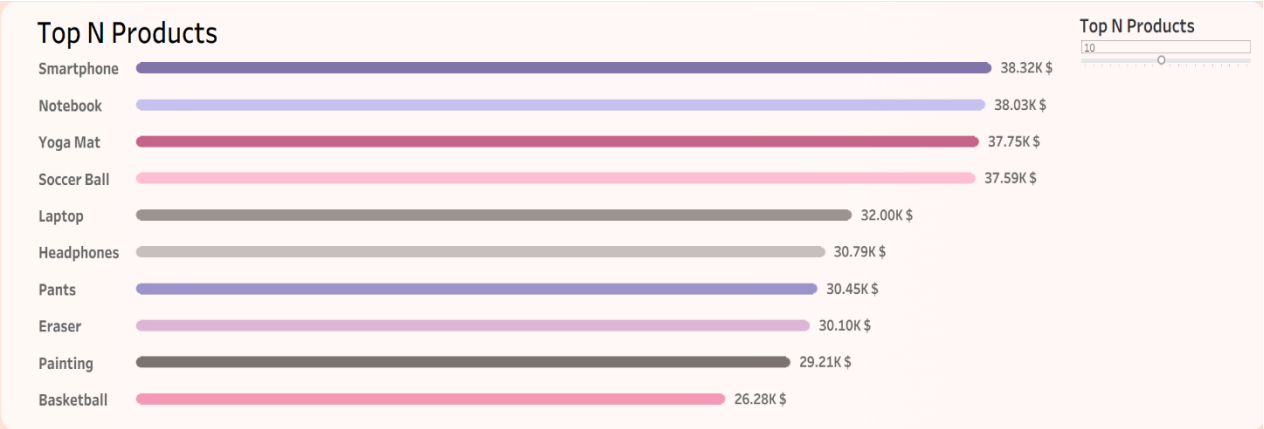


Figure 11: *Top Revenue Generating Products*

Outcome from Revenue dashboard:

Revenue is not a black box. It can be influenced by a small set of levers like category mix, product focus, payment pathways, and time or seasonality. These levers are actionable, but must be balanced against customer experience which is addressed in the third dashboard.

2.4.3 Dashboard 3: Customer Satisfaction

Purpose: To convert review ratings into measuring business risk, and to find ways to decrease low ratings at the operational level without having a negative impact on revenue.

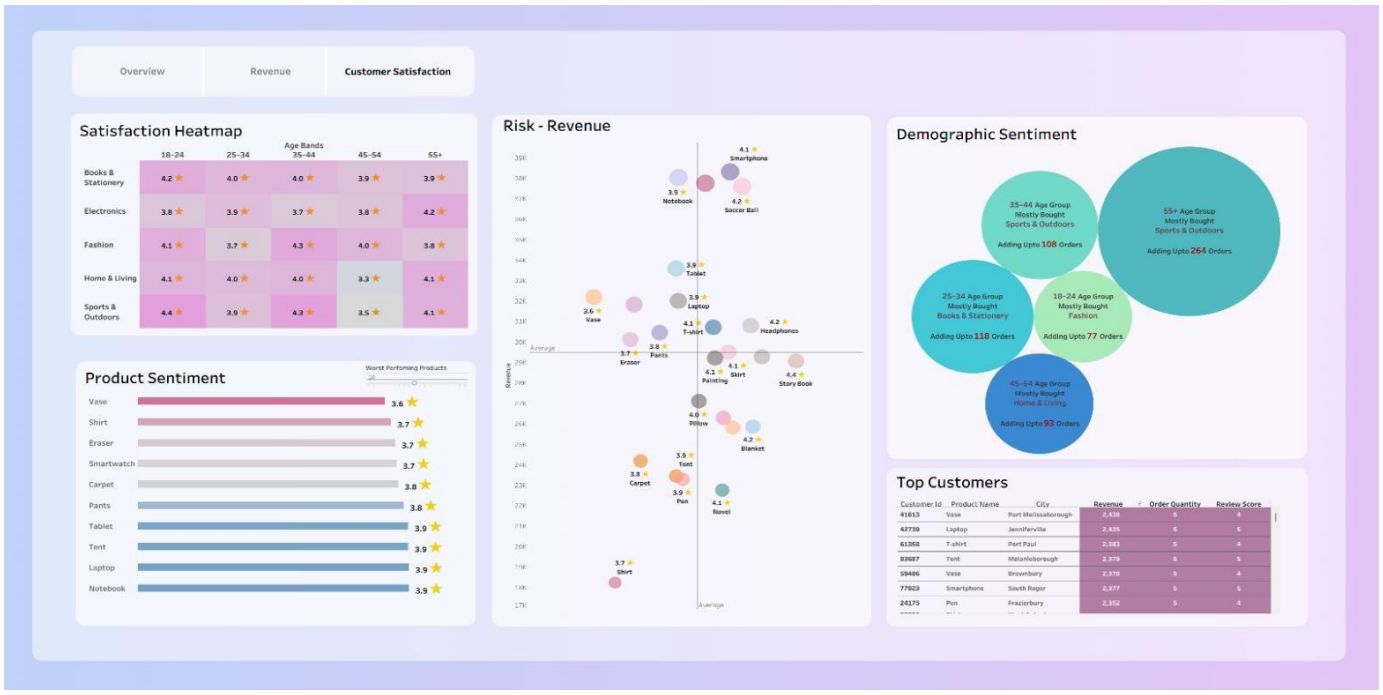
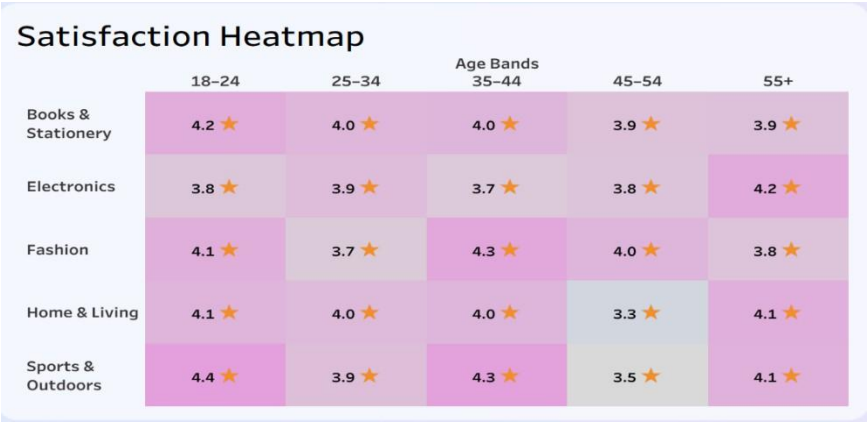


Figure 12: *Dashboard 3*

- **Satisfaction heatmap (Category × Age band):** this is the strongest diagnostic view for "who is unhappy and about what." It directly supports targeted action:
    - If a category is rated lower by specific age groups, then the response can be targeted for specific product fit, custom messaging, evaluating expectations and providing targeted support.
    - If a category underperforms across all ages, it signals broader product or service issues.



Satisfaction Heatmap

	Age Bands 18–24	25–34	35–44	45–54	55+
Books & Stationery	4.2 ★	4.0 ★	4.0 ★	3.9 ★	3.9 ★
Electronics	3.8 ★	3.9 ★	3.7 ★	3.8 ★	4.2 ★
Fashion	4.1 ★	3.7 ★	4.3 ★	4.0 ★	3.8 ★
Home & Living	4.1 ★	4.0 ★	4.0 ★	3.3 ★	4.1 ★
Sports & Outdoors	4.4 ★	3.9 ★	4.3 ★	3.5 ★	4.1 ★

Figure 13: *Customer Satisfaction with Age Demographics*

- **Risk–Revenue scatter plot (product-level positioning):** reframes ratings as a business trade-off. The chart helps prioritize:
    - High revenue + lower rating → Require urgent fixes as it protect revenue engine.
    - Low revenue + low rating → Deprioritize or discontinue the product.
    - High revenue + high rating → Safe revenue engine.

    This is the key device that connects experience to commercial impact.

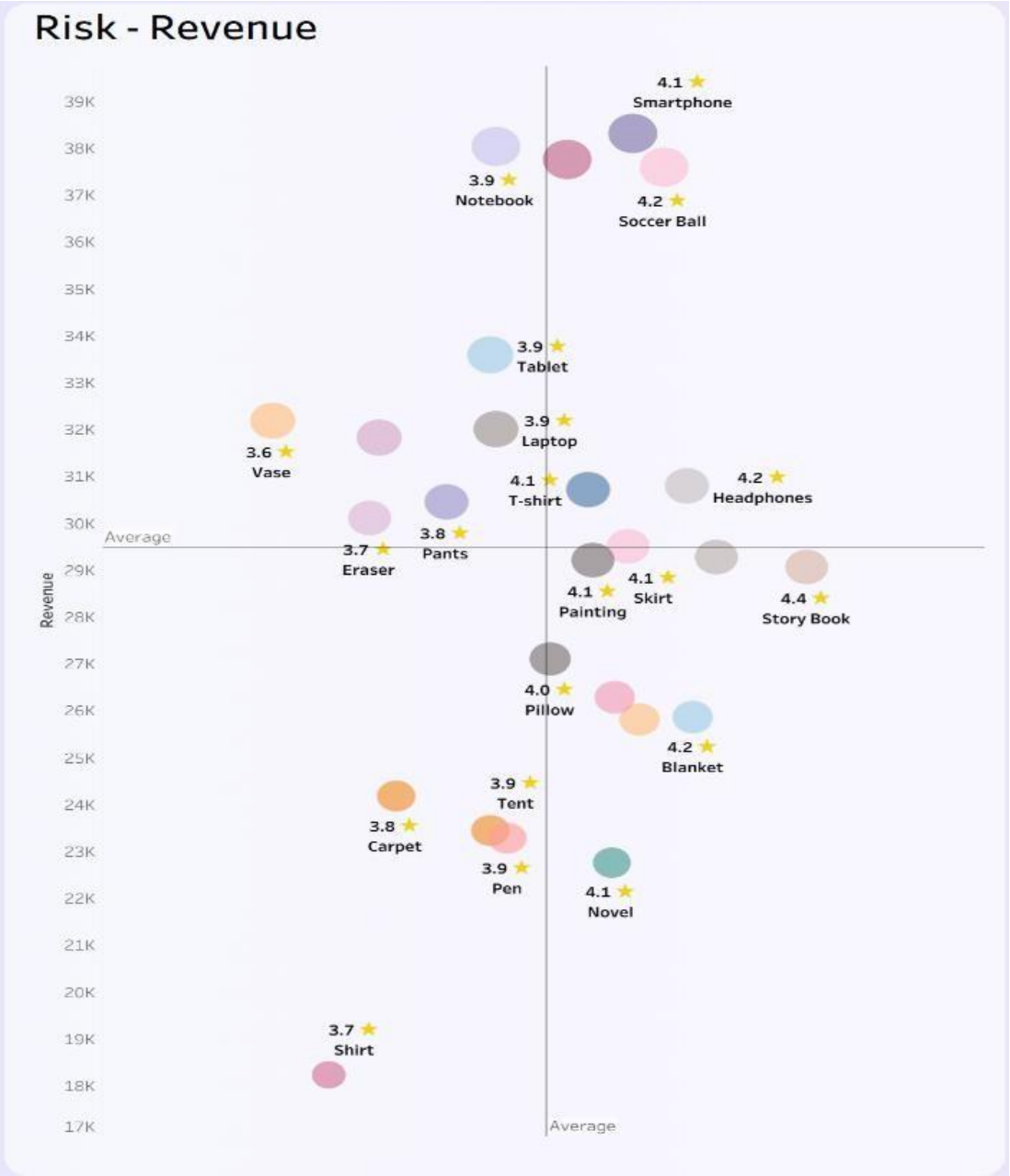


Figure 14: *Revenue Performance Compared with Risk Involved*

- **Product Sentiment:** Isolates products pulling ratings down. This makes the satisfaction view operational. It creates a concrete list for quality checks, product page improvement, or supplier review.

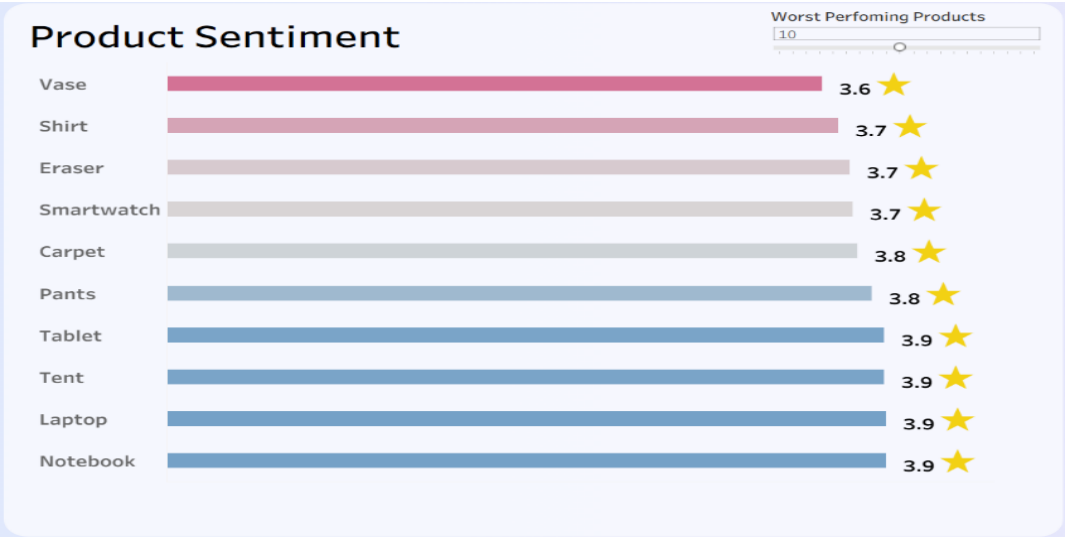


Figure 15: *Customer Sentiments on Products*

- **Demographic sentiment bubbles:** Summarizes which age groups dominate order activity and which categories they lean toward. This supports segmentation strategy like tailored bundles, targeted recommendations or segment-specific messaging.

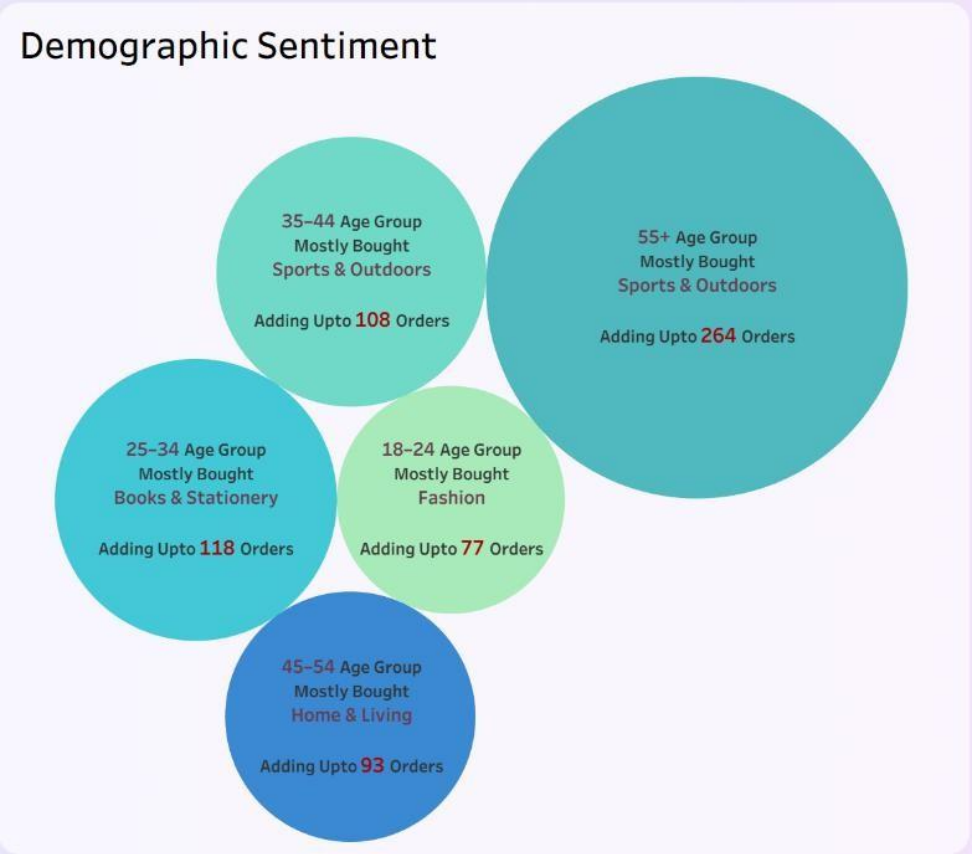


Figure 16: *Customer Sentiments and Sales*

- **Top Customers table:** included as a relationship cum retention lens. It allows quick identification of high-value customers and their purchasing pattern which are useful for loyalty initiatives and proactive services.

Top Customers

Customer Id	Product Name	City	Revenue	Order Quantity	Review Score
41613	Vase	Port Melissaborough	2,438	5	4
42739	Laptop	Jenniferville	2,435	5	5
61358	T-shirt	Port Paul	2,383	5	4
83687	Tent	Melanieborough	2,379	5	5
59486	Vase	Brownbury	2,378	5	4
77823	Smartphone	South Roger	2,377	5	5
24175	Pen	Frazierbury	2,352	5	4

Figure 17: *Top Customers in Revenue Contribution*

Outcome from Customer Satisfaction Dashboard:

Customer satisfaction is not uniform. Risk is concentrated in specific product or category and segment combinations. The risk–revenue view provides a rational way to prioritize fixes based on business impact.

2.5 Interactivity and Analyst Workflow Features:

To move beyond static reporting and meet the expectation of interactive, decision-oriented analytics, the dashboards were designed with:

- **Navigation across dashboards** (Overview → Revenue → Customer Satisfaction) to mirror a business conversation flow.

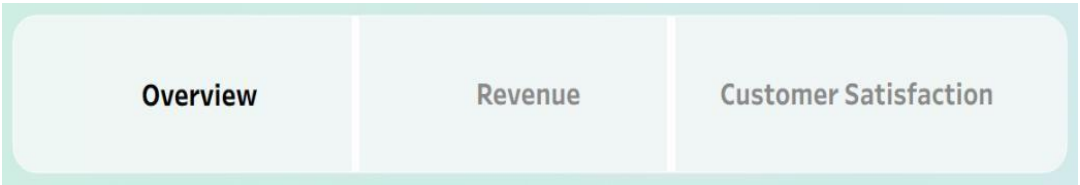


Figure 18: *Interactive Navigation Bar*

- **Top N parameters** for cities, products and worst performing products, so stakeholders can dynamically adjust depth without rebuilding charts.

- **Time controls** like month and quarter, so the same dashboard supports both tactical and strategic review.

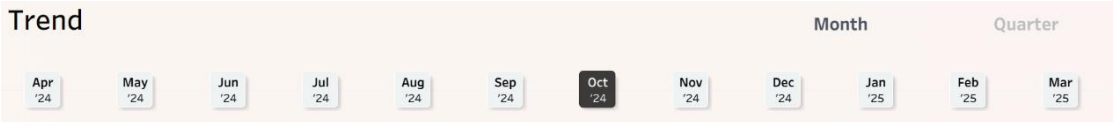


Figure 19: *Interactive Time Control for In Depth Analysis*

- **Cross-filtering potential** by which users can answer follow-up questions in real time.

3. Discussion

3.1 From Insight to Decision

The purpose of this analysis was not only descriptive reporting but also to translate patterns in revenue and customer satisfaction into practical business guidance. With the mix of performance metrics and experience indicators, the dashboards reveal interconnection between growth opportunities and customer rather than independence.

Three themes consistently emerge:

1) Revenue is structurally driven, not random.
2) Customer satisfaction is uneven across segments and products.
3) Growth decisions carry hidden experience risks.

3.2 Revenue Implications:

3.2.1 Diversified Revenue Base:

The category contribution view indicates that revenue is distributed across several major categories rather than dominated by a single segment. From a managerial perspective, this reduces concentration risk and suggests multiple growth pathways:

- Scaling already strong categories
- Revitalising mid-tier categories with upward trends
- Defending high-revenue categories showing stagnation or decline

Managerial implication:

Growth strategy should favour portfolio optimisation rather than over-reliance on a single best category.

3.2.2 Product-level leverage

The Top Products analysis highlights that a relatively small subset of products contributes disproportionately to revenue. This reflects a typical retail pattern where revenue engines coexist with long-tail items.

Managerial implication:

- Protect and continuously monitor high-performing products.
- Ensure availability, pricing competitiveness, and content quality.
- Use high-performing products as anchors for bundling and cross-sell.

3.2.3 Geographic Concentration

The Top Cities chart shows that certain cities contribute significantly more revenue.

Managerial implication:

- Prioritise logistics reliability and delivery service level agreements (SLAs) in high revenue cities.
- Deploy location-targeted promotions.
- Consider regional assortment adjustments.

3.2.4 Payment Behaviour as a Revenue Lever

The Payment Influence visualization suggests that revenue and order volume vary by payment method.

Managerial implication:

- Optimise checkout experience for dominant payment methods.
- Negotiate lower transaction costs for high-volume methods.
- Use payment-specific incentives strategically.

3.3 Customer Satisfaction Implications:

3.3.1 Overall Satisfaction Masks Variability

While the average rating (~4.0) indicates generally positive sentiment, the Satisfaction Heatmap reveals category × age band differences.

Managerial implication:

Customer experience strategies should be segment-specific, not based solely on aggregate averages. For examples:

- Categories rated lower by younger segments, improve product content, trend alignment.
- Categories rated lower by older segments shows usability, clarity and trust signals

3.3.2 Low-rating Risk Concentration

The Risk–Revenue scatterplot demonstrates that dissatisfaction risk is not evenly spread:

- Some products combine high revenue and lower ratings.
- Others show low revenue along with low ratings

Managerial implication:

Prioritisation should follow business impact logic:

  - High revenue + lower rating → Immediate intervention
  - Low revenue + low rating → Evaluate discontinuation
  - High revenue + high rating → Scale confidently

3.3.3 Worst-performing Products

The Worst Products view isolates items disproportionately associated with lower ratings.

Managerial implication:

- Conduct product quality review and supplier assessment.
- Improve descriptions, images and specifications on the selling platforms.
- Monitor return & refund patterns if possible.

3.4 Growth vs Experience Trade-off

A key managerial tension emerges: Short-term revenue optimisation can increase long-term satisfaction risk

Examples:

- Aggressive discounting may drive sales but lower perceived value.

- Expanding into weaker categories leads to revenue growth but creates rating volatility.
- Scaling problematic products amplifies dissatisfaction.

Managerial implication:

Decision-makers should adopt a balanced scorecard mindset of revenue growth, profitability, customer satisfaction and experience risk.

3.5 Business Operational Insights

Annual revenue depends almost entirely on August and December. Revenue trends (Figure 3) show these two months carry the budget while the rest of the year stays flat. Failing during these peak windows ruins the entire yearly performance. There is no room for operational mistakes when the financial stakes are this concentrated in just a few weeks.

981 cities make up all sales in the calendar year. One single location does not dominate the market and the top five cities combined bring in very little of the total revenue (Figure 5). Top cities like Port Melisdaborough and Patricaville may bring in higher revenue compared to the rest, but the top five cities combined bring in very little when compared to the total revenue. This makes shipping difficult. The company manages thousands of tiny deliveries to almost all cites instead of sending bulk orders to a few main hubs. This scattered network makes delivery slow and expensive, which causes the inconsistent review scores seen across the country.

Additionally, Seniors over age 55 are the most important customers, but the business is frustrating them. The Satisfaction Heatmap (Figure 13) shows these high spenders are the unhappiest when buying Fashion or Books. Payment methods are also split evenly between Bank Transfer, Credit Card, and Cash on Delivery (Figure 10). Having three equal payment methods is not a benefit; it is an operational mess. When a payment split is perfectly even, it proves the business has not made any single method fast or easy. Managing three separate systems forces the company to maintain three different technical setups, three different sets of fees, and three different ways of tracking money. This complexity makes the system more likely to glitch or slow down during busy times.

Therefore, Seniors are not frustrated by having options. They are frustrated by friction. If a senior is already unhappy because their Fashion or Book order is complicated or late, a clunky checkout is the final straw. For this high-spending group, any extra step, like waiting for a Bank

Transfer to clear or dealing with the hassle of Cash on Delivery is highly perceptible cause to abandon the brand entirely.

Moreover, Service falls behind as sales volume increases. Electronics makes the most money but has poor customer satisfaction compared to smaller categories. Moving high-priced items through a scattered delivery network trades quality for volume. This damages the brand's reputation and will eventually stop new growth. The business is currently prioritizing short-term sales over keeping its best customers happy.

3.6 Strategic Recommendations

Based on the integrated analysis:

- **Protect revenue engines**: Prioritise operational excellence for top products or categories.
- **Segment-aware experience management**: Align product mix, communication, and user experience with dominant age or gender segments.
- **Risk-informed growth strategy**: Use the Risk–Revenue matrix before scaling campaigns.
- **Payment optimisation strategy**: Focus on ease of payment and incentivised high-value payment channels.
- **Product portfolio refinement**: Address persistently low-rated products or rationalise assortment.

3.7 Concluding Perspective

The analysis demonstrates that revenue performance and customer satisfaction are interdependent strategic variables. Dashboards integrating both dimensions allow management to:

- Grow sustainably rather than opportunistically.
- Detect early experience risks.
- Allocate resources based on value and risk.

In essence, the retailer’s competitive advantage lies not only in selling more, but in selling better without eroding customer trust.

4. Limitations

While our dashboards offer a clear view of business health, several data limitations affect the reliability and validity of our findings. The most important issue is the absence of a unique order or invoice identifier in the dataset. Without this specific ID, we cannot group products into "baskets" or accurately measure how many items a customer buys in a single transaction. This makes our calculation of the average order value (AOV) more of an estimate rather than a solid fact, which limits our understanding of true customer behaviour. Additionally, the dataset provides the selling price but completely leaves out the product cost or any information on profit margins. This is a major limitation because a product that generates high revenue might have a very low profit margin, meaning it is not as important to the business as it seems.

Moreover, the lack data on promotions, discounts, or coupon usage leads to not being able to tell if a sudden jump in sales was due to a clever marketing campaign or if people only bought the product because it was heavily discounted. Furthermore, we do not have an order status field to track if an item was successfully delivered or cancelled. Finally, the dataset is synthetic and small, with only 1,000 rows. This "idealized" data lacks the real-world complexity and messy errors found in actual retail environments. Without larger, multi-year data, it is difficult to prove that the patterns we see, like seasonal peaks or age-based preferences would stay the same in a real, fluctuating market.

To improve the suitability of this analysis and provide more actionable steps, we would need to include behavioural and operational details that are currently missing. For example, we have star ratings to show customer satisfaction, but we lack the actual review text or timestamps for those reviews. This means we can identify which categories are underperforming, but we cannot explain the specific reasons why it is happening. We do not know if customers are complaining about product quality, poor packaging, or bad value for money. If we have information such as, delivery service data, such as shipping dates and carrier performance, this will allow us to see if low satisfaction scores in certain cities are caused by slow delivery rather than the products themselves. Another critical gap is the lack of return and refund records. Since returns are the clearest sign of a failed transaction, our current "risk" assessment is incomplete without them. Furthermore, our geographic data is limited to city names only; having postcodes or latitude coordinates would allow for much more precise mapping of demand. For future work, we would also look for deeper customer profiles, such as loyalty tiers and acquisition channels, to see how long-term customers behave differently from new ones.

By including marketing costs and shipping fees, we could transform this from a revenue-focused report into a complete strategy for profitable and sustainable growth.

5. Conclusion

In summary, our analysis shows that the business is currently in a very strong financial position, with a total revenue of approximately $737,000 and a solid average customer rating of 4.0 stars across 1,000 customers. We found that money is coming in from several different areas like Electronics and Fashion, which is a good sign because it means the company is not relying on just one type of product to survive. However, the most important discovery was the "Risk-Revenue" trade-off. While many items are doing well, some of the top-selling "hero products" have lower-than-average ratings. This is a hidden danger because if these high-earning items continue to disappoint customers, it could hurt the brand's reputation and lead to a drop in future sales. The main lesson for management is that while the business is growing, they must focus on "selling better" by protecting their top products and fixing the quality issues in specific cities and age groups.

Looking ahead, there are several ways to expand this research to get an even clearer picture of how the business is really performing. Since we currently only have revenue data, a key next step would be to include product costs and profit margins. This would allow the company to see which items are making the most money after expenses, rather than just looking at total sales. We also recommend adding delivery service logs and the actual text from customer reviews. Having this information would help us understand exactly why customers are unhappy, whether it is because of slow shipping or poor product quality, so that the company can take the right actions to fix the problem. Finally, including data on returns and refunds would give us a much more accurate view of customer loyalty and the true financial risk the business faces in the long run.

6. Additional Work

This section lists the background work and data preparation:

- Data Engineering: Before building the charts, we created calculated fields in Tableau to turn raw data into useful metrics. This included calculating Total Revenue (Quantity * Price) and using COUNTD to find the exact number of unique customers.
- KPI Baseline: We established baseline metrics for Average Order Value (AOV) at $737 to help compare different customer segments and cities against the company average.

- Parameter Controls: We built Top N parameters into the “Revenue” and “Customer Satisfaction” dashboards. This allows a user to dynamically switch between seeing the Top 10 or Top 20 products and cities without needing a new report.
- Exploratory Data Analysis (EDA): We performed initial checks on monthly seasonality to confirm that the revenue peaks were consistent and not just caused by a single outlier month.

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Appendix

Dataset Link: Online Retail C E-commerce Dataset

Tableau Dashboard (Resolution: 3840x2160): E-commerce Dashboard (Group-6).twbx